

EE 133

Chapter 1. Crystal Properties and Growth of Semiconductors

Streetman and Banerjee, 'Solid State Electronic Devices,' 6th ed., Pearson, 2006.

1

Crystals and Semiconductors

- As EEs, we are interested in the electrical properties of solids.
 - The electrical properties depend on the type of atoms and their arrangement (crystalline, amorphous, etc.)
- Semiconductors – materials having electrical conductivity between metals and insulators. Electrical conductivity can be varied exponentially with temperature, impurity doping, or electrical gating.

2

Semiconductors

- Elemental semiconductors - Column IV
 - Si and Ge
- Binary semiconductors – III-V or II-VI
 - GaAs, AlAs, InP, InAs, InSb, AlSb, GaSb, GaP, AlP, GaN
 - ZnS, ZnSe, ZnTe, ZnO, CdS, CdSe, CdTe,

Periodic table of the elements

Legend:

- alkali metals
- alkaline earth metals
- transition metals
- other metals
- halogens
- noble gases
- rare earth elements (21, 39, 57–71)
- lanthanide elements (57–71 only)
- actinide elements

* Numbering system adopted by the International Union of Pure and Applied Chemistry (IUPAC).
 ** Numbering system widely used, especially in the U.S., from the mid-20th century.
 *** Discoveries of elements 113–116 and 118 are claimed but not confirmed. Element names and symbols in parentheses are temporarily assigned by IUPAC.

© 2009 Encyclopædia Britannica, Inc.

3

Semiconductors

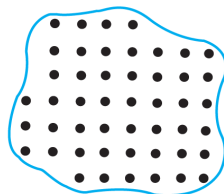
- Si is the dominant semiconductor material. Used for all electronic applications. Mixed with Ge.
 - cpu, RAM, integrated circuits (IC) chips.
- III-Vs and II-VIs are used for optoelectronic applications
 - LEDs: GaN, GaP, GaAs
 - LASERS: InGaAsP (fiber optic communication systems).

4

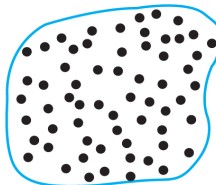
Semiconductor Atomic Structure

All semiconductors are crystals.

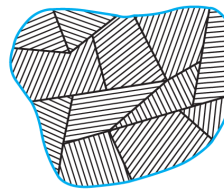
Crystal is a periodic structure – a periodic arrangement of atoms.



Crystalline
Si, Ge, etc.



Amorphous
(random)
Glass
 SiO_2



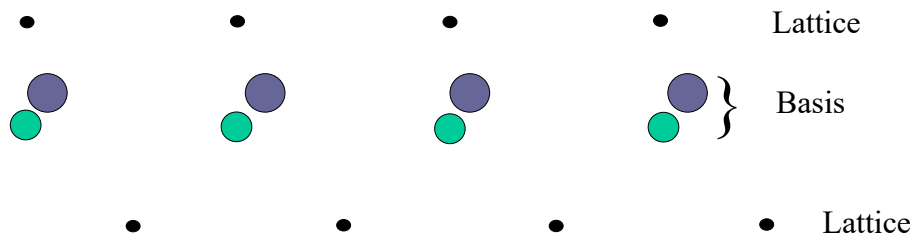
Poly crystalline
Adjacent single crystal grains

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

5

Crystal Structure Concepts

- Lattice – Periodic array of points in space that define the periodicity of the crystal (Bravais Lattice).
- Basis – Atoms associated with each lattice point.
- 1D example:

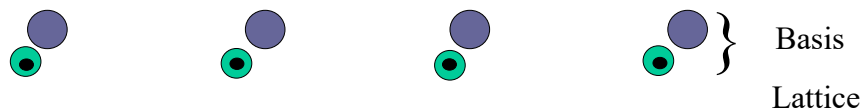


Lattice is not unique – only the periodicity matters

6

Crystal Structure Concepts

- Basis – Usually one of the basis atoms is placed directly on a lattice point – but not necessary.

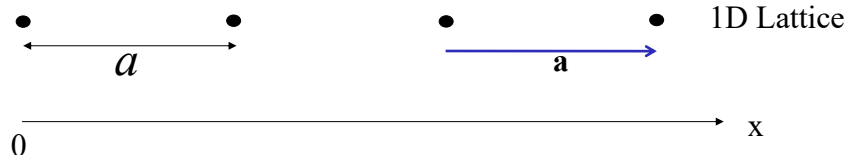


7

Crystal Structure Concepts

- Primitive lattice vector: $\mathbf{a} = a \hat{\mathbf{x}}$
– A vector which defines the spatial period of the lattice.

1D Example:



For any physical property of the crystal $f(\mathbf{r})$,

$$f(\mathbf{r}) = f(\mathbf{r} + n\mathbf{a}) \quad (n = \text{integer})$$

i.e. the function is periodic with spatial period $\mathbf{a}$.

e.g. Electron density.

Analogous to a periodic time signal:

$$f(t) = f(t + n\tau)$$

8

Crystal Structure Concepts

- Primitive lattice vectors: $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$

- In 3D, there are 3 primitive lattice vectors, $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.

$$f(\mathbf{r}) = f(\mathbf{r} + n\mathbf{a} + m\mathbf{b} + p\mathbf{c}) \quad (n, m, p = \text{integers})$$

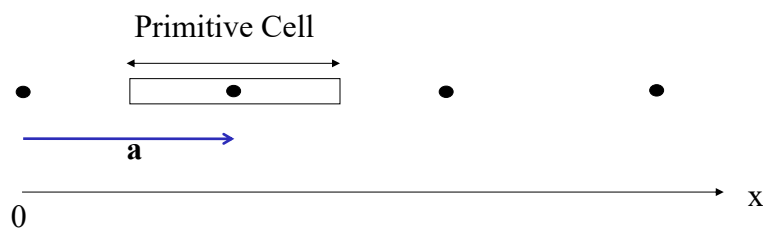
- Any point $\mathbf{r} = \mathbf{r}_0 + n\mathbf{a} + m\mathbf{b} + p\mathbf{c}$ is indistinguishable from $\mathbf{r}_0$ (n, m, p integers).

9

Crystal Structure Concepts

- Primitive cell: Smallest repeat unit of the lattice.

- Infinite lattice is created by repeating the primitive cell by integer multiples of the primitive lattice vectors.
- One lattice point per primitive cell.



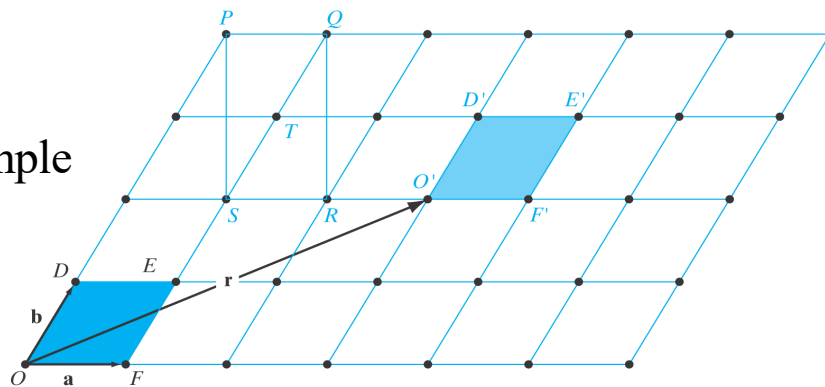
10

Crystal Structure Concepts

- Unit Cell: A cell, larger than the primitive cell, which is conveniently chosen to be cubic for the semiconductors that we discuss (Si, Ge, GaAs, etc.)

11

• 2D Example

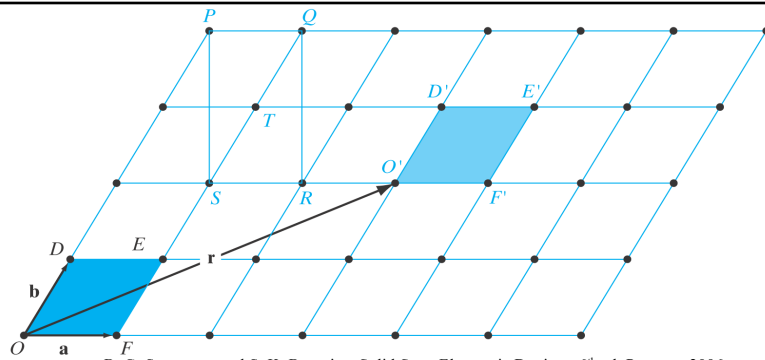


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

1. All lattice points are equivalent.
2. A lattice point P at $\mathbf{R}_p$ is given by $\mathbf{R}_p = m\mathbf{a} + n\mathbf{b}$ where (m,n integers).
e.g. $\mathbf{R}_S = \mathbf{a} + 2\mathbf{b}$; $\mathbf{r} = \mathbf{R}_{Q'} = 3\mathbf{a} + 2\mathbf{b}$
3. Primitive Cell: ODEF.
 1. Entire lattice is obtained by repeating ODEF by $m\mathbf{a} + n\mathbf{b}$.
 2. Primitive cell is smallest cell possible which can be repeated by $m\mathbf{a} + n\mathbf{b}$ to create the lattice (no interior lattice points).
 3. Effective no. of points = 1.

12

Unit Cell



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

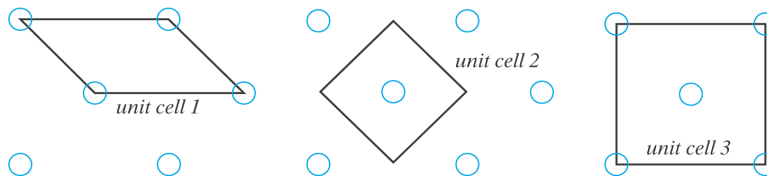
Unit Cell: PQRS

- Larger than the primitive cell with a simpler, rectangular or cubic geometry.
- 1 interior point T
- 1 exterior point: $(P+Q+R+S) / 4$
 $\rightarrow$ 2 lattice points / rectangular unit cell.
- Lattice is created by translation of unit cell by its lattice vectors: $\{a\hat{x}, \sqrt{3}b\hat{y}\}$
- Unit cell lattice vectors aligned along x, y, z $\rightarrow$ orthogonal. Primitive lattice vectors *not* orthogonal.

13

Example

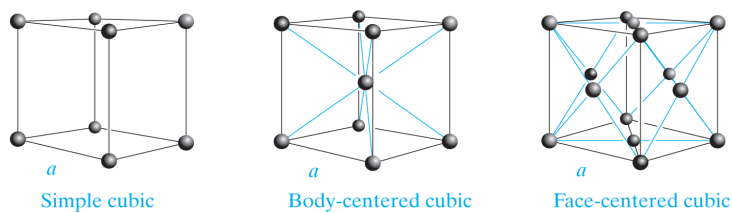
- Each of the blue circles is a lattice point. Which of the cells below are primitive cells? Why?



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

14

- Unit Cell: A cell, larger than the primitive cell, which is conveniently chosen to be cubic for the semiconductors that we discuss (Si, Ge, GaAs, etc.)

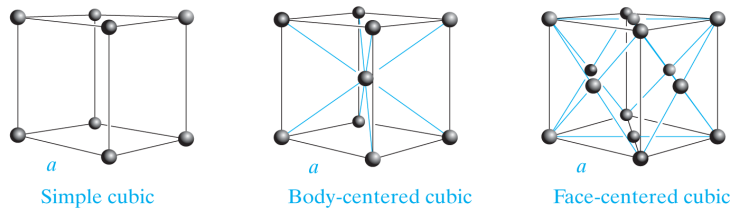


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

15

3D Lattices

- Bravais showed that there are only 14 different ways of arranging points in space to make 3D lattices. 14 Bravais lattices and corresponding unit cells.



Simple Cubic (SC)
Lattice point at each
corner of cell

Body-Centered Cubic
(BCC)
Lattice point at each corner
of cell + 1 atom at center

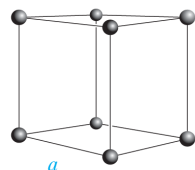
Face Centered Cubic (FCC)
Lattice point at each corner +
1 atom at center of each face

Unit cell lattice vectors: $\mathbf{a} = a\hat{\mathbf{x}}, \mathbf{b} = a\hat{\mathbf{y}}, \mathbf{c} = a\hat{\mathbf{z}}$

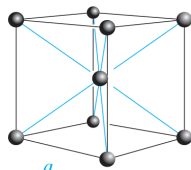
B. G. Streetman and S. K. Banerjee,
Solid State Electronic Devices, 6th ed.
Pearson, 2006.

16

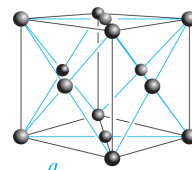
3D Lattices



Simple cubic



Body-centered cubic

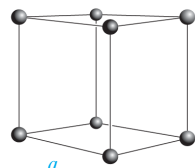


Face-centered cubic

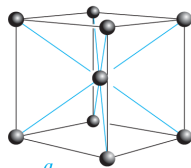
B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

- Assume one atom on each lattice point.
- What is the nearest neighbor distance for each cubic unit cell?
 - SC: a
 - BCC: $\left| \frac{a}{2}(\hat{x} + \hat{y} + \hat{z}) \right| = a \frac{\sqrt{3}}{2}$
 - FCC: $\left| \frac{a}{2}(\hat{x} + \hat{y}) \right| = a \frac{\sqrt{2}}{2}$

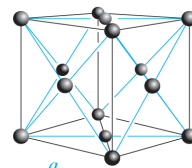
17



Simple cubic



Body-centered cubic

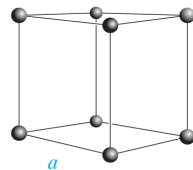


Face-centered cubic

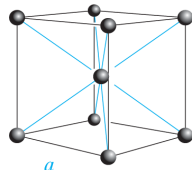
B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

- Assume one atom on each lattice point.
- How many atoms are associated with each unit cell?
 - SC: Each atom is shared by 8 cells $\rightarrow$ 1/8 atom per corner times 8 corners = 1 atom.
 - BCC: 8 corners $\rightarrow$ 1 atom
+ 1 atom at center = 2 atoms.
 - FCC: 8 corners $\rightarrow$ 1 atom
Each face atom is shared by 2 cells $\rightarrow$ 1/2 atom per face times 6 faces = 3 atoms.
Total: Corners (1) + Faces (3) = 4 atoms.

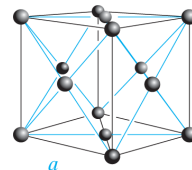
18



Simple cubic



Body-centered cubic



Face-centered cubic

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

- Assume one atom on each lattice point.
- How many nearest neighbors does each atom have?

– SC:

– BCC:

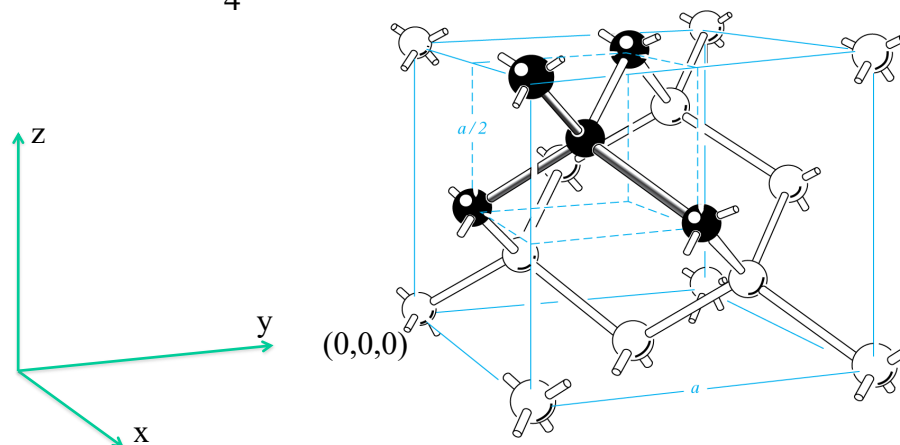
– FCC:



19

Diamond Structure: Si, Ge, C

- FCC lattice with 2 atoms per lattice point.
 - 1 atom at each lattice point
 - 1 shifted by $\frac{a}{4}(\hat{x} + \hat{y} + \hat{z})$

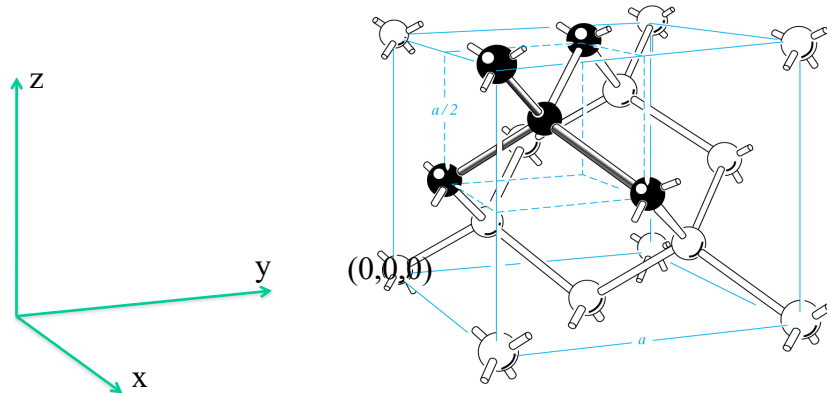


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

20

Zinc Blende Structure: III-V GaAs, InAs, GaP, InP, etc.

- FCC lattice with 2 **different** atoms per lattice point.
 - 1 atom of type A at each lattice point
 - 1 atom of type B shifted by $\frac{a}{4}(\hat{x} + \hat{y} + \hat{z})$

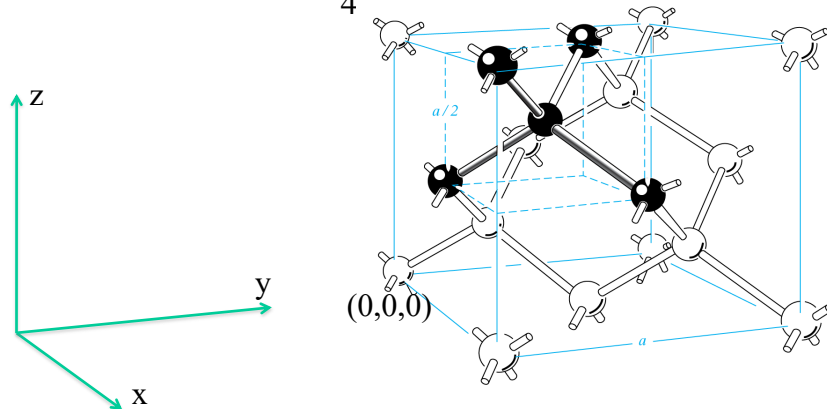


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

21

Zinc Blende Structure: III-V GaAs, InAs, GaP, InP, etc.

- Diamond structure where the 2 atoms are different.
 - The atom on the FCC lattice is Ga
 - The atom shifted by $\frac{a}{4}(\hat{x} + \hat{y} + \hat{z})$ is As.

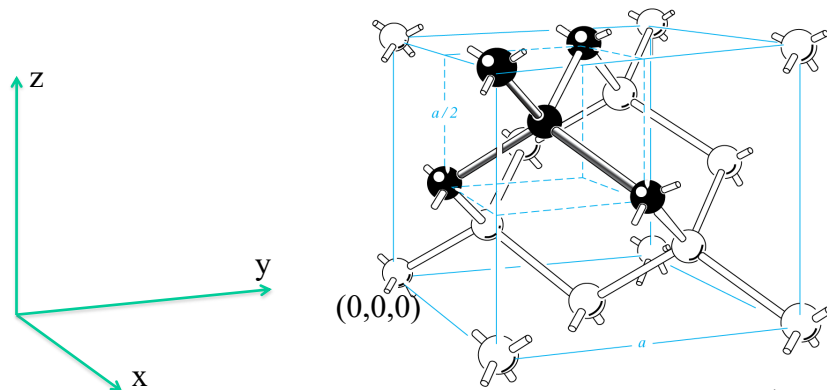


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

22

Diamond or Zinc Blende Structure:

- Described as 2 interpenetrating FCC lattices.
 - The first (Ga) start at (0,0,0).
 - The second (As) FCC sub-lattice is displaced by $\frac{a}{4}(\hat{x} + \hat{y} + \hat{z})$ from the first.



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

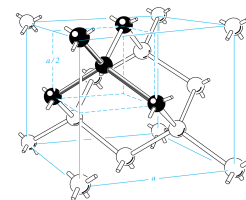
23

Diamond or Zinc Blende Structure:

- Number of atoms in unit cell?
FCC + 4 interior atoms = 8 atoms.
- What is the density of Si given that $a = 5.43 \text{ \AA}$ and the atomic weight of Si = 28.09 g/mol?

Idea: (Weight of the 8 atoms in the unit cell) / (volume of the unit cell)

$$\rho = \frac{28.09 \text{ (g/mol)} \times 8 \text{ atoms}}{\left(6.02 \times 10^{23} \frac{\text{atoms}}{\text{mol}}\right) \left(5.43 \text{ \AA}\right)^3 \left(\frac{1 \text{ cm}}{10^8 \text{ \AA}}\right)^3} = 2.33 \frac{\text{g}}{\text{cm}^3}$$



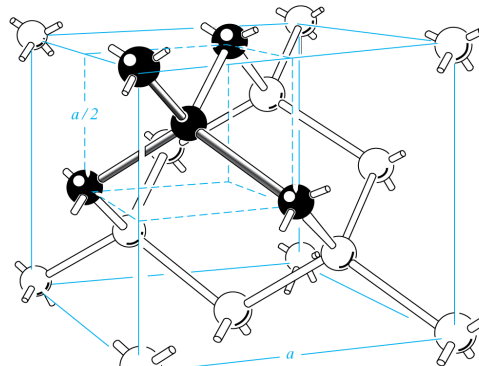
B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

24

Diamond or Zinc Blende Structure:

- Number of nearest neighbor atoms for diamond or zinc blende?

4 nearest neighbors. Each Si atom bonds to 4 others.



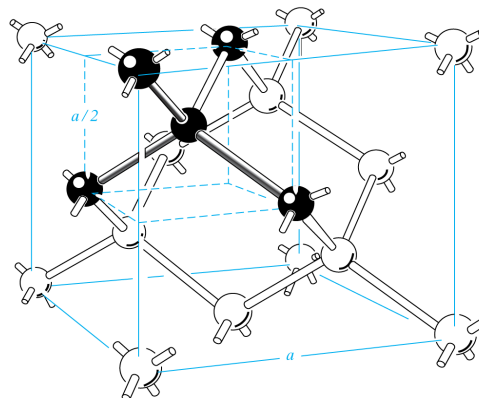
B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

25

Diamond or Zinc Blende Structure:

- Nearest neighbor bond length for diamond or zinc blende?

$$\left| \frac{a}{4}(\hat{x} + \hat{y} + \hat{z}) \right| = a \frac{\sqrt{3}}{4}$$



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

26

Review

- Semiconductors
 - IV: Si, Ge, C
 - III-V: GaAs, InP, InAs, GaP, etc.
 - II-VI: ZnS, CdS, ZnSe, CdSe, ZnTe, CdTe
- Crystals
 - Lattice: points that represent the periodicity of crystal
 - Basis: atoms associated with each lattice point
 - Primitive cell: Smallest repeat unit
 - Cubic unit cell: SC, BCC, FCC, diamond, zinc-blende
 - In diamond or zinc-blende, there are 8 atoms / unit cell.
 - Each atom has 4 nearest neighbors.
 - 1 atom on each FCC lattice pt. and another offset by $\frac{a}{4}(\hat{x} + \hat{y} + \hat{z})$

27

New Topics

- Crystal directions.
- Crystal planes.
 - Miller Indices
- Equivalent directions and planes.
- Si crystal growth technology.

28

Crystal Directions

- Need to describe directions of a crystal.
- Set up a coordinate system corresponding to the $\hat{x}$, $\hat{y}$, and $\hat{z}$ axes of the cubic unit cell.
- Directions are specified in terms of a triplet in square brackets. The numbers are always **integers**.
 - $\hat{x}$ direction $\rightarrow [100]$
 - $\hat{y}$ direction $\rightarrow [010]$
 - $\hat{z}$ direction $\rightarrow [001]$
 - $\hat{x} + \hat{y}$ direction $\rightarrow [110]$
 - $\hat{x} + \hat{y} + 3\hat{z}$ direction $\rightarrow [113]$
 - $\hat{x} + 1.5\hat{y}$ direction $\rightarrow [230]$

Negative Directions: Indicate with a bar.

$$-\hat{x} \Rightarrow [\bar{1}00]$$

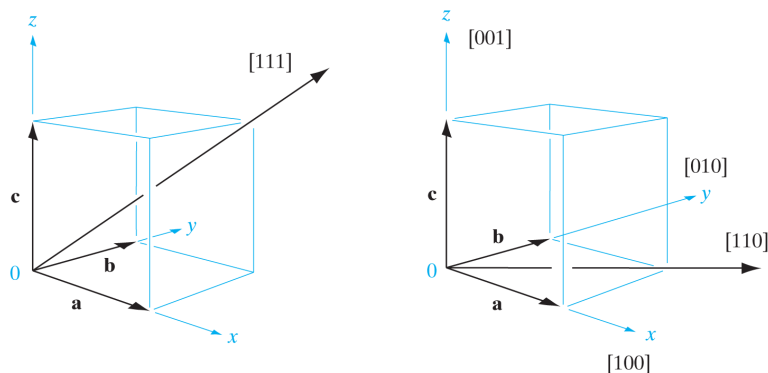
$$\hat{x} - \hat{y} \Rightarrow [1\bar{1}0]$$

$$\hat{x} + \hat{y} - \hat{z} \Rightarrow [11\bar{1}]$$

29

Crystal Directions

- Illustration



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

30

Equivalent Directions

- Because of symmetry, many directions are equivalent. i.e. What you see looking in the $[100]$ direction looks the same as the $[\bar{1}00]$, $[010]$, $[0\bar{1}0]$, $[001]$, and $[00\bar{1}]$ directions.
 - All of these directions are **equivalent** and depend only on an arbitrary choice of axes.
 - This equivalent set of directions is indicated by angled brackets:
$$\langle 100 \rangle$$
- $[100] \rightarrow$ x direction.
- $\langle 100 \rangle \rightarrow$ equivalent set of $[100]$ directions.

31

Equivalent Directions

- Example: Write down all directions belonging to $\langle 110 \rangle$.
- $[110]$, $[\bar{1}10]$, $[1\bar{1}0]$
- $[101]$, $[\bar{1}01]$, $[10\bar{1}]$
- $[011]$, $[0\bar{1}1]$, $[01\bar{1}]$

32

Crystal Planes

- For cubic unit cells, the plane is indicated by its normal direction in (). i.e. the plane normal to $[100]$ is the (100) plane.
 - Plane normal to $[110]$ is the (110) plane.
- **Equivalent planes**: $\{100\}$ is the set of equivalent (100) planes.
 - (100) , (010) , (001) , $(\bar{1}00)$, $(0\bar{1}0)$, $(00\bar{1})$
- **Miller Indices**: (h, k, ℓ) are called the Miller indices.

33

Miller Indices

- General recipe to find Miller Indices of a plane.
- Find intercepts of plane with crystal axis. Allowed to rigidly shift planes to avoid origin. Express as integer multiples of unit cell vectors.
- Take reciprocal of the 3 integers and multiply by the lowest common denominator to again get 3 integers, h , k , and ℓ .
- Plane is $(h k \ell)$.

34

Miller Indices

Example:

- Plane with intercepts at $2\mathbf{a}$, $3\mathbf{b}$, $4\mathbf{c}$.
 - $(1/2, 1/3, 1/4) \times 12 = (6\ 4\ 3)$ plane.
- x-y plane:
 - Shift up by $\mathbf{c}$.
 - Intercepts at $\infty\mathbf{a}$, $\infty\mathbf{b}$, $1\mathbf{c}$
 - Reciprocals: $(0\ 0\ 1)$ plane.
- Exception: Intercept is a fraction of a lattice constant. e.g. a plane perpendicular to $[100]$ going through the body center atom of a BCC cell.
 - Intercepts: $\frac{1}{2}\mathbf{a}$, $\infty\mathbf{b}$, $\infty\mathbf{c} \rightarrow (2\ 0\ 0)$ plane.

35

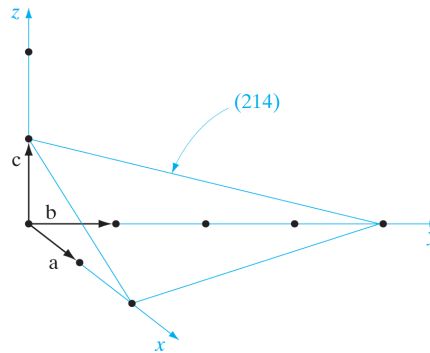
Common Planes and Directions

- Common planes and directions are
 $\{100\}$, $\{111\}$, $\{110\}$, $\langle 100 \rangle$, $\langle 111 \rangle$, $\langle 110 \rangle$

36

Miller Indices Example

- Intercepts are $2a$, $4b$, $1c$
- Reciprocals are $(1/2, 1/4, 1)$
- Multiply by 4: $(2\ 1\ 4)$ plane.

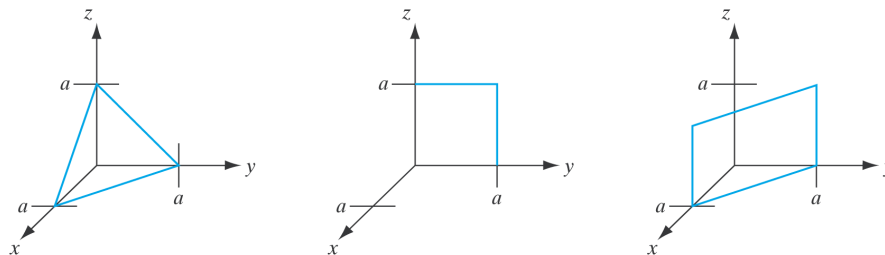


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

37

Miller Indices Example

- What are the Miller Indices for each plane below?

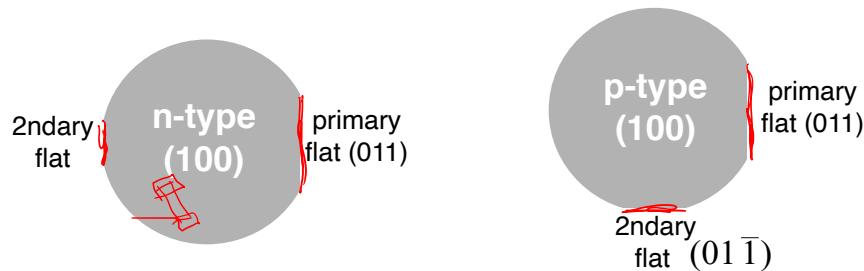


B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

38

Miller Indices in Everyday Use

- Si Wafers
- Different diameters: 1 in, 2 in, 3 in, 4 in, 6 in, 8 in, 12 in.
- Different orientations:
 - Most common: (100) wafer, i.e. the top surface (plane) is (100).
 - Also order (111) wafers. Top surface is (111).
- Orientation flats on (100) wafers:



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

39

Growing Bulk Si

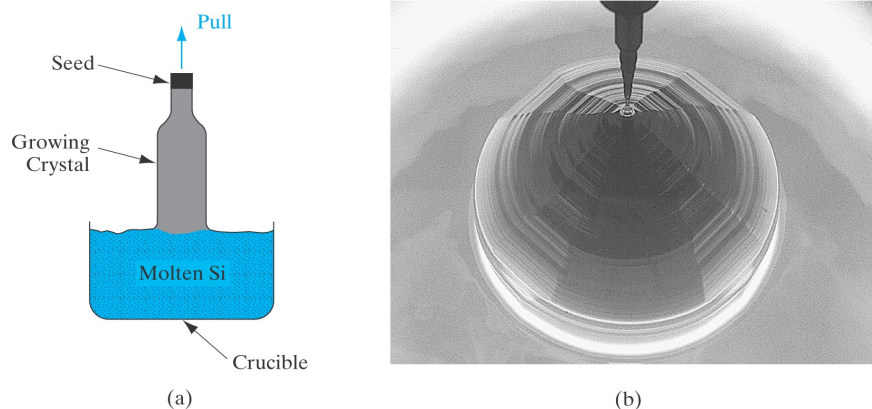


Figure 1.10

Pulling of a Si crystal from the melt (Czochralski method): (a) schematic diagram of the crystal growth process; (b) an 8-in. diameter, (100)-oriented Si crystal being pulled from the melt.

(Photograph courtesy of MEMC Electronics Intl.)

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

40

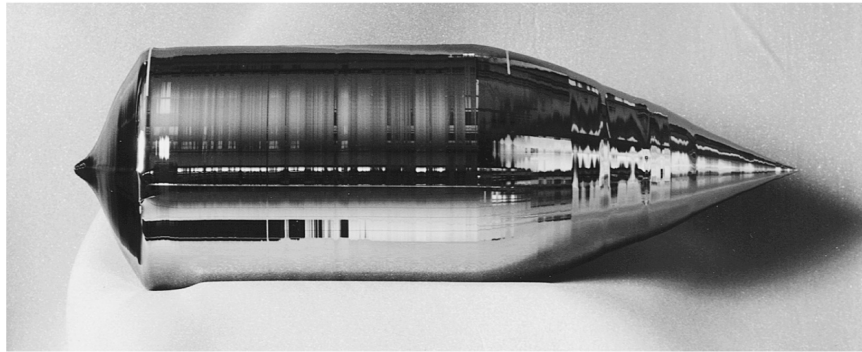


Figure 1.11

Silicon crystal grown by the Czochralski method. This large single-crystal ingot provides 300 mm (12-in.) diameter wafers when sliced using a saw. The ingot is about 1.0 m long (including the tapered regions), and weighs about 140 kg. (Photograph courtesy of MEMC Electronics Intl.)

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

41



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

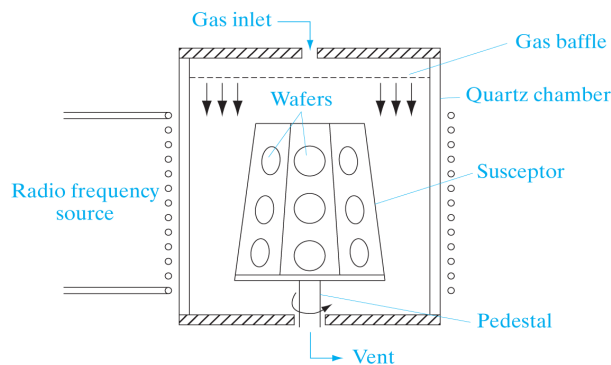
42

Epitaxial Growth

- Grow a thin (1 nm – 100 nm) crystal layer on a wafer substrate.
 - Chemical vapor deposition (CVD) or Vapor phase epitaxy (VPE). Used in Si IC process. High throughput, thick layers.
 - Molecular Beam Epitaxy (MBE): Thin, high quality layers with sharp interfaces. (Jianlin Liu's lab)

43

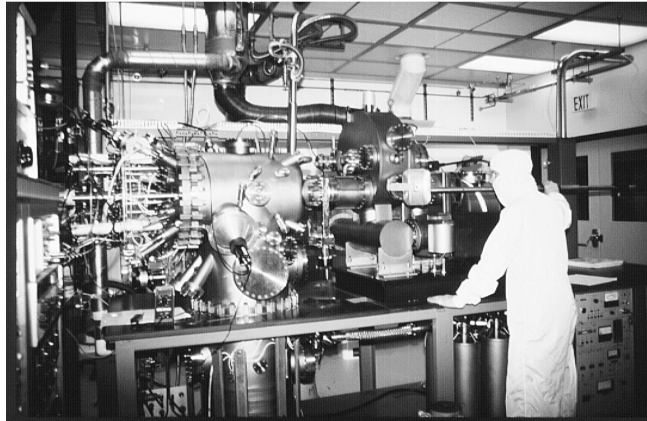
CVD or VPE



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

44

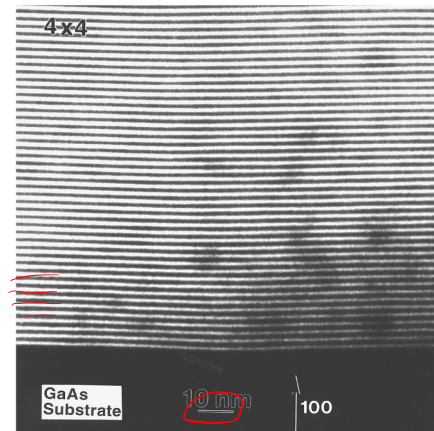
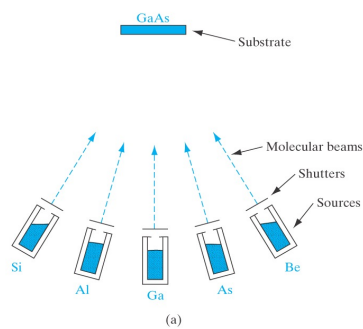
MBE



B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

45

MBE



TEM of wafer

B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

46

Summary

- Semiconductors
 - IV: S, Ge, C
 - III-V: GaAs, InP, InAs, GaP, etc.
 - II-VI: ZnS, CdS, ZnSe, CdSe, ZnTe, CdTe
- Crystals
 - Lattice: points that represent the periodicity of crystal
 - Basis: atoms associated with each lattice point
 - Primitive cell: Smallest repeat unit
 - Cubic unit cell: SC, BCC, FCC, diamond, zinc-blende
 - Directions: $[100]$, $[\bar{1}00]$, $\langle 100 \rangle$
 - Planes: (100) , $\{100\}$ – Miller indices ($h \ k \ \ell$)

47

Summary (cont.)

- Crystals (cont.)
 - Si wafer production Czochralski growth
 - Epitaxial growth: VPE or CVD and MBE

48